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The Accuracy of Percentages

In solving a problem of this sort, the grand thing is to be able to reason backward. That is a very useful accomplishment, and a very easy one, but people do not practise it much . . . . Most people, if you describe a train of events to them, will tell you what the result would be. They can put those events together in their minds, and argue from them that something will come to pass. There are few people, however, who, if you told them a result, would be able to evolve from their own inner consciousness what the steps were which led up to that result. This power is what I mean when I talk of reasoning backward . . . .

—*Sherlock Holmes*[1]

1. INTRODUCTION

The previous chapter reasoned from the box to the draws. Draws were made at random from a box whose composition was known, and a typical problem was finding the chance that the percentage of 1's among the draws would be in a given interval. As Sherlock Holmes points out, it is often very useful to turn this reasoning around, going instead from the draws to the box. A statistician would call this *inference* from the sample to the population. Inference is the topic of this chapter.

For example, suppose a survey organization wants to know the percentage of Democrats in a certain district. They might estimate it by taking a simple random sample. Naturally, the percentage of Democrats in the sample would be used to estimate the percentage of Democrats in the district—an example of reasoning backward from the draws to the box. Because the sample was chosen at random,

it is possible to say how accurate the estimate is likely to be, just from the size and composition of the sample. This chapter will explain how.

The technique is one of the key ideas in statistical theory. It will be presented in the polling context. A political candidate wants to enter a primary in a district with 100,000 eligible voters, but only if he has a good chance of winning. He hires a survey organization, which takes a simple random sample of 2,500 voters. In the sample, 1,328 favor the candidate, so the percentage is

$$\frac{1{,}328}{2{,}500} \times 100\% \approx 53\%.$$

The candidate is discussing this result with his pollster.

Politician. I win.

Pollster. Not so fast. You want to know the percentage you'd get among all the voters in the district. We only have it in the sample.

Politician. But with a good sample, it's bound to be the same.

Pollster. Not true. It's what I said before. The percentage you get in the sample is different from what you'd get in the whole district. The difference is what we call chance error.

Politician. Could the sample be off by as much as three percentage points? If so, I lose.

Pollster. Actually, we can be about 95% confident that we're right to within two percentage points. It looks good.

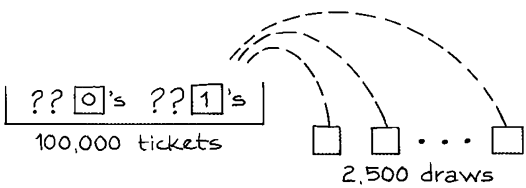
"I'M BEHIND YOU 100 PERCENT, PLUS OR MINUS 3 PERCENT OR SO."

Politician. What gives you the size of the chance error?

Pollster. The standard error. Remember, we talked about that the other day. As I was telling you . . . .

Politician. Sorry, I'm expecting a phone call now.

The politician has arrived at the crucial question to ask when considering survey data: how far wrong is the estimate likely to be? As the pollster wanted to say, the likely size of the chance error is given by the standard error. To figure that, a box model is needed. There should be one ticket in the box for each voter, making 100,000 tickets in all. Each ticket should be marked 1 or 0, where 1 means a vote for the candidate, 0 a vote against him. There are 2,500 draws made at random from the box. The data are like the draws, and the number of voters in the sample who favor the candidate is like the sum of the draws. This completes the model.



To get the SE for the sum, the survey organization needs the SD of the box. This is

$$\sqrt{(\text{fraction of 1's}) \times (\text{fraction of 0's})}.$$

At this point, the pollsters seem to be stuck. They don't know how each ticket in the box should be marked. They don't even know the fraction of 1's in the box. That parameter represents the fraction of voters in the district who favor their candidate, which is exactly what they were hired to find out. (Hence the question marks in the box.)

Survey organizations lift themselves over this sort of obstacle by their own bootstraps.[2] They substitute the fractions observed in the sample for the unknown fractions in the box. In the example, 1,328 people out of the sample of 2,500 favored the candidate. So $1{,}328/2{,}500 \approx 0.53$ of the sample favored him, and the other 0.47 were opposed. The estimate is that about 0.53 of the 100,000 tickets in the box are marked 1, the other 0.47 being marked 0.

On this basis, the SD of the box is estimated as $\sqrt{0.53 \times 0.47} \approx 0.50$. The SE for the number of voters in the sample who favor the candidate is estimated as $\sqrt{2{,}500} \times 0.50 = 25$. The 25 measures the likely size of the chance error in the 1,328. Now 25 people out of 2,500 (the size of the sample) is 1%. The SE for the percentage of voters in the sample favoring the candidate is estimated as 1 percentage point. This completes the bootstrap procedure for estimating the standard error.

As far as the candidate is concerned, this calculation shows that his pollster's estimate of 53% is only likely to be off by 1 percentage point or so. It is very

unlikely to be off by as much as 3 percentage points—that's 3 SEs. He is well on the safe side of 50%, and he should enter the primary.

> *The bootstrap*. When sampling from a 0–1 box whose composition is unknown, the SD of the box can be estimated by substituting the fractions of 0's and 1's in the sample for the unknown fractions in the box. The estimate is good when the sample is reasonably large.

The bootstrap procedure may seem crude. But even with moderate-sized samples, the fraction of 1's among the draws is likely to be quite close to the fraction in the box. Similarly for the 0's. If survey organizations use their sample fractions in the formula for the SD of the box, they are not likely to be far wrong in estimating the SE.

One point is worth more discussion. The expected value for the number of 1's among the draws (translation—the expected number of sample voters who favor the candidate) is

$$2{,}500 \times \text{fraction of 1's in the box.}$$

This is unknown, because the fractions of 1's in the box is unknown. The SE of 25 says about how far the 1,328 is from its expected value. In statistical terminology, the 1,328 is an observed value; the contrast is with the unknown expected value. (Observed values are discussed on p. 292.)

Example 1. In fall 2005, a city university had 25,000 registered students. To estimate the percentage who were living at home, a simple random sample of 400 students was drawn. It turned out that 317 of them were living at home. Estimate the percentage of students at the university who were living at home in fall 2005. Attach a standard error to the estimate.

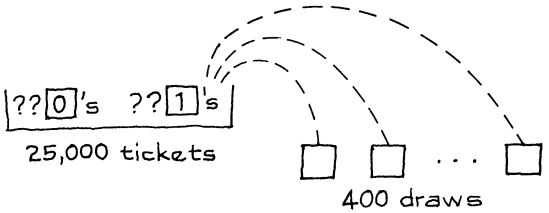
Solution. The sample percentage is

$$\frac{317}{400} \times 100\% \approx 79\%$$

That is the estimate for the population percentage.

For the standard error, a box model is needed. There are 25,000 tickets in the box, one for each student in the population. There are 400 draws from the box, one for each student in the sample. This problem involves classifying and counting, so each ticket in the box should be marked 1 or 0. We are counting students who were living at home. The tickets corresponding to these students should be marked 1; the others, 0. There are 400 draws made at random from the box. The data are like the draws, and the number of students in the sample who were living at home is like the sum of the draws. That completes the model. (See the sketch at the top of the next page.)

The fraction of 1's in the box is a parameter. It represents the fraction of all the students at this university who were living at home in fall 2005. It is unknown, but can be estimated as 0.79—the fraction observed in the sample. Similarly, the



fraction of 0's in the box is estimated as 0.21. So the SD of the box is estimated by the bootstrap method as $\sqrt{0.79 \times 0.21} \approx 0.41$. The SE for the number of students in the sample who were living at home is estimated as $\sqrt{400} \times 0.41 \approx 8$. The 8 gives the likely size of the chance error in the 317. Now convert to percent, relative to the size of the sample:

$$\frac{8}{400} \times 100\% = 2\%$$

The SE for the sample percentage is estimated as 2%. Let's summarize. In the sample, 79% of the students were living at home. The 79% is off the mark by 2 percentage points or so. That is what the SE tells us.

The discussion in this section focused on simple random sampling, where the mathematics is easiest. In practice, survey organizations use much more complicated designs. Even so, with probability methods it is generally possible to say how big the chance errors are likely to be—one of the great advantages of probability methods for drawing samples.

Exercise Set A

1. Fill in the blanks, and explain.
   (a) In example 1 on p. 378, the 317 is the _____ value for the number of students in the sample who were living at home. Options:
      (i) expected (ii) observed
   (b) The SD of the box is _____ 0.41. Options:
      (i) exactly equal to (ii) estimated from the data as
   (c) The SE for the number of students in the sample who were living at home is _____ 8. Options: (i) exactly equal to (ii) estimated from the data as

2. In a certain city, there are 100,000 persons age 18 to 24. A simple random sample of 500 such persons is drawn, of whom 194 turn out to be currently enrolled in college. Estimate the percentage of all persons age 18 to 24 in that city who are currently enrolled in college.[3] Put a give-or-take number on the estimate.
   (a) The first step in solving this problem is:
      (i) finding the SD of the box.
      (ii) finding the average of the box.
      (iii) writing down the box model.
      Choose one option, and explain.
   (b) Now solve the problem.

3. In a simple random sample of 100 graduates from a certain college, 48 were earning $50,000 a year or more. Estimate the percentage of all graduates of that college earning $50,000 a year or more.[4] Put a give-or-take number on the estimate.

4. A simple random sample of size 400 was taken from the population of all manufacturing establishments in a certain state: 11 establishments in the sample had 100 employees or more. Estimate the percentage of manufacturing establishments with 100 employees or more.[5] Attach a standard error to the estimate.

5. In the same state, a simple random sample of size 400 was taken from the population of all persons employed by manufacturing establishments: 187 people in the sample worked for establishments with 100 employees or more. Estimate the percentage of people who worked for establishments with 100 employees or more. Attach a standard error to the estimate.

6. Is the difference between the percentages in exercises 4 and 5 due to chance error?

The next two exercises are designed to illustrate the bootstrap method for estimating the SD of the box.

7. Suppose there is a box of 100,000 tickets, each marked 0 or 1. Suppose that in fact, 20% of the tickets in the box are 1's. Calculate the standard error for the percentage of 1's in 400 draws from the box.

8. Three different people take simple random samples of size 400 from the box in exercise 7, without knowing its contents. The number of 1's in the first sample is 72. In the second, it is 84. In the third, it is 98. Each person estimates the SE by the bootstrap method.
   (a) The first person estimates the percentage of 1's in the box as _____, and figures this estimate is likely to be off by _____ or so.
   (b) The second person estimates the percentage of 1's in the box as _____, and figures this estimate is likely to be off by _____ or so.
   (c) The third person estimates the percentage of 1's in the box as _____, and figures this estimate is likely to be off by _____ or so.

9. In a certain town, there are 25,000 people aged 18 and over. To estimate the percentage of them who watched a certain TV show, a statistician chooses a simple random sample of size 1,000. As it turns out, 308 of the sample people did see the show. Complete the following table; the first 3 lines refer to the sample percentage who saw the show. (N/A = not applicable.)

	Known to be	*Estimated from the data as*
Observed value	30.8%	N/A
Expected value	N/A	30.8%
SE		
SD of box		
Number of draws		

The answers to these exercises are on pp. A81–82.

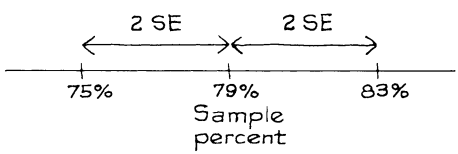
2. CONFIDENCE INTERVALS

In the example of the previous section, 79% of the students in the sample were living at home: the sample percentage was 79%. How far can the population percentage be from 79%? (Remember, "population percentage" means the percentage of all students at the university who were living at home.) The standard error was estimated as 2%, suggesting a chance error of around 2% in size. So the population percentage could easily be 77%. This would mean a chance error of 2%:

sample percentage	=	population percentage	+	chance error
79%	=	77%	+	2%

The population percentage could also be 76%, corresponding to a chance error of 3%. This is getting unlikely, because 3% represents 1.5 SEs. The population percentage could even be as small as 75%, but this is still more unlikely; 4% represents 2 SEs. Of course, the population percentage could be on the other side of the sample percentage, corresponding to negative chance errors. For instance, the population percentage could be 83%. Then the estimate is low by 4%: the chance error is −4%, which is −2 SEs.

With chance errors, there is no sharp dividing line between the possible and the impossible. Errors larger in size than 2 SEs do occur—infrequently. What happens with a cutoff at 2 SEs? Take the interval from 2 SEs below the sample percentage to 2 SEs above:



This is a *confidence interval* for the population percentage, with a *confidence level* of about 95%. You can be about 95% confident that the population percentage is caught inside the interval from 75% to 83%.

What if you want a different confidence level? Anything except 100% is possible, by going the right number of SEs in either direction from the sample percentage. For instance:

- The interval "sample percentage ± 1 SE" is a 68%-confidence interval for the population percentage.
- The interval "sample percentage ± 2 SEs" is a 95%-confidence interval for the population percentage.
- The interval "sample percentage ± 3 SEs" is a 99.7%-confidence interval for the population percentage.

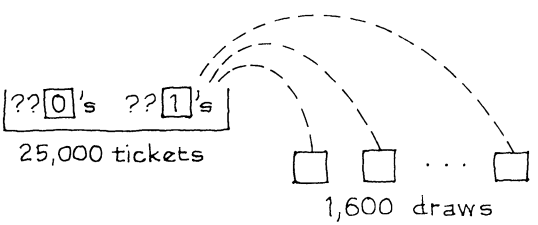
However, even 10 SEs may not give 100% confidence, because there is the remote possibility of very large chance errors. There are no definite limits to the normal curve: no matter how large a finite interval you choose, the normal curve has some area outside that interval.[6]

Example 2. A simple random sample of 1,600 persons is taken to estimate the percentage of Democrats among the 25,000 eligible voters in a certain town. It turns out that 917 people in the sample are Democrats. Find a 95%-confidence interval for the percentage of Democrats among all 25,000 eligible voters.

Solution. The percentage of Democrats in the sample is

$$\frac{917}{1{,}600} \times 100\% \approx 57.3\%.$$

The estimate: about 57.3% of the eligible voters in the town are Democrats. For the standard error, a box model is needed. There is one ticket in the box for each eligible voter in the town, making 25,000 tickets in all. There are 1,600 draws, corresponding to the sample size of 1,600. This problem involves classifying (Democrat or not) and counting, so each ticket is marked 1 or 0. It is Democrats that are being counted. So the tickets corresponding to Democrats are marked 1, the others are marked 0. There are 1,600 draws made at random from the box. The data are like the draws, and the number of Democrats in the sample is like the sum of the draws. That completes the model.



The fraction of 1's in the box (translation—the fraction of Democrats among the 25,000 eligible voters) is unknown, but can be estimated by 0.573, the fraction of Democrats in the sample. Similarly, the fraction of 0's in the box is estimated as 0.427. So the SD of the box is estimated by the bootstrap method as $\sqrt{0.573 \times 0.427} \approx 0.5$. The SE for the number of Democrats in the sample is estimated as $\sqrt{1{,}600} \times 0.5 = 20$. The 20 gives the likely size of the chance error in the 917. Now convert to percent, relative to the size of the sample:

$$\frac{20}{1{,}600} \times 100\% = 1.25\%.$$

The SE for the percentage of Democrats in the sample is 1.25%. The percentage of Democrats in the sample is likely to be off the percentage of Democrats in the population, by 1.25 percentage points or so. A 95%-confidence interval for the percentage of Democrats among all 25,000 eligible voters is

$$57.3\% \pm 2 \times 1.25\%.$$

That is the answer. We can be about 95% confident that between 54.8% and 59.8% of the eligible voters in this town are Democrats.

Confidence levels are often quoted as being "about" so much. There are two reasons. (i) The standard errors have been estimated from the data. (ii) The nor-

mal approximation has been used. If the normal approximation does not apply, neither do the methods of this chapter. There is no hard-and-fast rule for deciding. The best way to proceed is to imagine that the population has the same percentage composition as the sample. Then try to decide whether the normal approximation would work for the sum of the draws from the box. For instance, a sample percentage near 0% or 100% suggests that the box is lopsided, so a large number of draws will be needed before the normal approximation takes over (section 5 of chapter 18). On the other hand, if the sample percentage is near 50%, the normal approximation should be satisfactory when there are only a hundred draws or so.

Exercise Set B

1. Fill in the blanks, and explain.
   (a) In example 2 on p. 382, the 917 is the _____ value for the number of Democrats in the sample. Options: (i) expected (ii) observed
   (b) The SD of the box is _____ $\sqrt{0.573 \times 0.427}$. Options:
   (i) exactly equal to (ii) estimated from the data as
   (c) The SE for the number of Democrats in the sample is _____ 20. Options:
   (i) exactly equal to (ii) estimated from the data as

2. Refer back to exercise 2 on p. 379.
   (a) Find a 95%-confidence interval for the percentage of persons age 18 to 24 in the city who are currently enrolled in college.
   (b) Repeat, for a confidence level of 99.7%.
   (c) Repeat, for a confidence level of 99.7%, supposing the size of the sample was 2,000, of whom 776 were currently enrolled in college.

3. A box contains 1 red marble and 99 blues; 100 marbles are drawn at random with replacement.
   (a) Find the expected number of red marbles among the draws, and the SE.
   (b) What is the chance of drawing fewer than 0 red marbles?
   (c) Use the normal curve to estimate this chance.
   (d) Does the probability histogram for the number of red marbles among the draws look like the normal curve?

4. A box contains 10,000 marbles, of which some are red and the others blue. To estimate the percentage of red marbles in the box, 100 are drawn at random without replacement. Among the draws, 1 turns out to be red. The percentage of red marbles in the box is estimated as 1%, with an SE of 1%. True or false: a 95%-confidence interval for the percentage of red marbles in the box is $1\% \pm 2\%$. Explain.

The answers to these exercises are on pp. A82–83.

3. INTERPRETING A CONFIDENCE INTERVAL

In example 1 on p. 378, a simple random sample was taken to estimate the percentage of students registered at a university in fall 2005 who were living at home. An approximate 95%-confidence interval for this percentage ran from 75%

to 83%, because

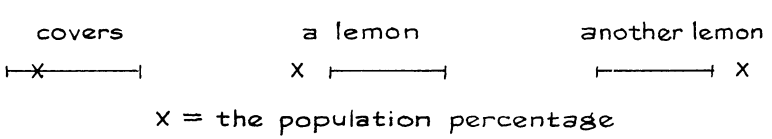
$$\text{sample percentage} \pm 2\,\text{SE} = 75\% \text{ to } 83\%.$$

It seems more natural to say "There is a 95% chance that the population percentage is between 75% and 83%." But there is a problem here. In the frequency theory, a chance represents the percentage of the time that something will happen. No matter how many times you take stock of all the students registered at that university in the fall of 2005, the percentage who were living at home back then will not change. Either this percentage was between 75% and 83%, or not. There really is no way to define the chance that the parameter will be in the interval from 75% to 83%. That is why statisticians have to turn the problem around slightly.[7] They realize that the chances are in the sampling procedure, not in the parameter. And they use the new word "confidence" to remind you of this.

The chances are in the sampling procedure, not in the parameter.

The confidence level of 95% says something about the sampling procedure, and we are going to see what that is. The first point to notice: the confidence interval depends on the sample. If the sample had come out differently, the confidence interval would have been different. With some samples, the interval "sample percentage ± 2 SE" traps the population percentage. (The word statisticians use is *cover*.) But with other samples, the interval fails to cover. It's like buying a used car. Sometimes you get a lemon—a confidence interval which doesn't cover the parameter.

Three confidence intervals



The confidence level of 95% can now be interpreted. For about 95% of all samples, the interval

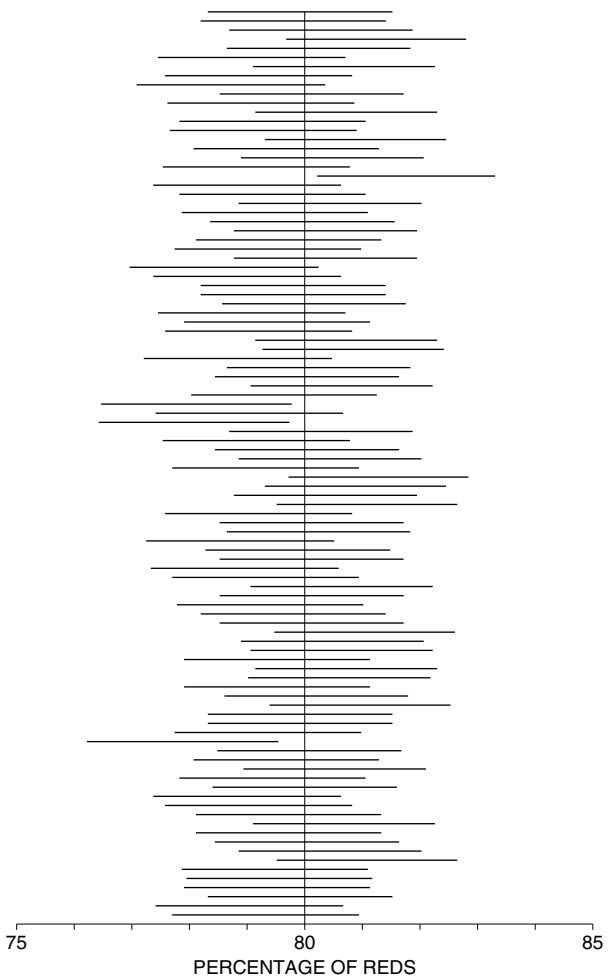
$$\text{sample percentage} \pm 2\,\text{SE}$$

covers the population percentage, and for the other 5% it does not. Of course, investigators usually cannot tell whether their particular interval covers the population percentage, because they do not know that parameter. But they are using a procedure that works 95% of the time: take a simple random sample, and go 2 SEs either way from the sample percentage. It is as if their interval was drawn at random from a box of intervals, where 95% cover the parameter and only 5% are lemons. This beats second-hand cars.

> A confidence interval is used when estimating an unknown parameter from sample data. The interval gives a range for the parameter, and a confidence level that the range covers the true value.

Confidence levels are a bit difficult, because they involve thinking not only about the actual sample but about other samples that could have been drawn. The interpretation is illustrated in figure 1. A hundred survey organizations are hired to estimate the percentage of red marbles in a large box. Unknown to the pollsters,

Figure 1. Interpreting confidence intervals. The 95%-confidence interval is shown for 100 different samples. The interval changes from sample to sample. For about 95% of the samples, the interval covers the population percentage, marked by a vertical line.[8]



this percentage is 80%. Each organization takes a simple random sample of 2,500 marbles, and computes a 95%-confidence interval for the percentage of reds in the box, using the formula

$$\text{percentage of reds in sample} \pm 2\,\text{SE}.$$

The percentage of reds is different from sample to sample, and so is the estimated standard error. As a result, the intervals have different centers and lengths. Some of the intervals cover the percentage of red marbles in the box, others fail. About 95% of them should cover the percentage, which is marked by a vertical line. In fact, 96 out of 100 do. Of course, this is only a computer simulation, designed to illustrate the theory. In practice, an investigator would have only one sample, and would not know the parameter.

Probabilities are used when you reason forward, from the box to the draws; confidence levels are used when reasoning backward, from the draws to the box (see the chapter opening quote). There is a lot to think about here, but keep the main idea of the chapter in mind.

> A sample percentage will be off the population percentage, due to chance error. The SE tells you the likely size of the amount off.

Confidence levels were introduced to make this idea more quantitative.

Exercise Set C

1. Probabilities are used when reasoning from the _____ to the _____; confidence levels are used when reasoning from the _____ to the _____. Options:
   box draws

2. (a) The chance error is in the _____ value. Options: observed, expected
   (b) The confidence interval is for the _____ percentage. Options:
   sample population

3. Refer to exercises 7 and 8 on p. 380. Compute a 95%-confidence interval for the percentage of 1's in the box, using the data obtained by the person in exercise 8(a). Repeat for the other two people. Which of the three intervals cover the population percentage, that is, the percentage of 1's in the box? Which do not? (Remember, the three people in exercise 8 do not know the contents of the box; but you do, from exercise 7.)

4. A box contains a large number of red and blue marbles; the proportion of red marbles is known to be 50%. A simple random sample of 100 marbles is drawn from the box. Say whether each of the following statements is true or false, and explain briefly.
   (a) The percentage of red marbles in the sample has an expected value of 50%, and an SE of 5%.
   (b) The 5% measures the likely size of the chance error in the 50%.

(c) The percentage of reds in the sample will be around 50%, give or take 5% or so.
(d) An approximate 95%-confidence interval for the percentage of reds in the sample is 40% to 60%.
(e) There is about a 95% chance that the percentage of reds in the sample will be in the range from 40% to 60%.

5. A box contains a large number of red and blue marbles, but the proportions are unknown; 100 marbles are drawn at random, and 53 turn out to be red. Say whether each of the following statements is true or false, and explain briefly.
   (a) The percentage of red marbles in the box can be estimated as 53%; the SE is 5%.
   (b) The 5% measures the likely size of the chance error in the 53%.
   (c) The 53% is likely to be off the percentage of red marbles in the box, by 5% or so.
   (d) A 95%-confidence interval for the percentage of red marbles in the box is 43% to 63%.
   (e) A 95%-confidence interval for the percentage of red marbles in the sample is 43% to 63%.

6. A simple random sample of 1,000 persons is taken to estimate the percentage of Democrats in a large population. It turns out that 543 of the people in the sample are Democrats. True or false, and explain:
   (a) The sample percentage is $(543/1{,}000) \times 100\% = 54.3\%$; the SE for the sample percentage is 1.6%.
   (b) $54.3\% \pm 3.2\%$ is a 95%-confidence interval for the population percentage.
   (c) $54.3\% \pm 3.2\%$ is a 95%-confidence interval for the sample percentage.
   (d) There is about a 95% chance for the percentage of Democrats in the population to be in the range $54.3\% \pm 3.2\%$.

7. (Continues exercise 6; hard.) True or false, and explain: If another survey organization takes a simple random sample of 1,000 persons, there is about a 95% chance that the percentage of Democrats in their sample will be in the range $54.3\% \pm 3.2\%$.

8. At a large university, 54.3% of the students are female and 45.7% are male. A simple random sample of 1,000 persons is drawn from this population. The SE for the sample percentage of females is figured as 1.6%. True or false: There is about a 95% chance for the percentage of females in the sample to be in the range $54.3\% \pm 3.2\%$. Explain.

The answers to these exercises are on pp. A83–84.

4. *CAVEAT EMPTOR*

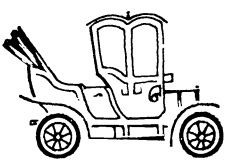
The methods of this chapter were developed for simple random samples. They may not apply to other kinds of samples. Many survey organizations use fairly complicated probability methods to draw their samples (section 4 of chapter 19). As a result, they have to use more complicated methods for estimating their standard errors. Some survey organizations do not bother to use probability methods at all. Watch out for them.

> Warning. The formulas for simple random samples may not apply to other kinds of samples.

Here is the reason. Logically, the procedures in this chapter all come out of the square root law (section 2 of chapter 17). When the size of the sample is small relative to the size of the population, taking a simple random sample is just about the same as drawing at random with replacement from a box—the basic situation to which the square root law applies. The phrase "at random" is used here in its technical sense: at each stage, every ticket in the box has to have an equal chance to be chosen. If the sample is not taken at random, the square root law does not apply, and may give silly answers.[9]

People often think that a statistical formula will somehow check itself while it is being used, to make sure that it applies. Nothing could be further from the truth. In statistics, as in old-fashioned capitalism, the responsibility is on the consumer.

Caveat emptor



Let the buyer beware

$$\bar{x} \pm z_\alpha \times s/\sqrt{n}$$

Exercise Set D

1. A psychologist is teaching a class with an enrollment of 100. He administers a test of passivity to these students and finds that 20 of them score over 50. The conclusion: approximately 20% of all students would score over 50 on this test. Recognizing that this estimate may be off a bit, he estimates the likely size of the error as follows:

$$\text{SE for number} = \sqrt{100} \times \sqrt{0.2 \times 0.8} = 4$$

$$\text{SE for percent} = (4/100) \times 100\% = 4\%$$

   What does statistical theory say?

2. A small undergraduate college has 1,000 students, evenly distributed among the four classes: freshman, sophomore, junior, and senior. In order to estimate the percentage of students who have ever smoked marijuana, a sample is taken by the following procedure: 25 students are selected at random without replacement from each of the four classes. As it turns out, 35 out of the 100 sample students admit to having smoked. So, it is estimated that 35% out of the 1,000 students at the college would admit to having smoked. A standard error is attached to this estimate, by the following procedure:

$$\text{SE for number} = \sqrt{100} \times \sqrt{0.35 \times 0.65} \approx 5$$

$$\text{SE for percent} = (5/100) \times 100\% = 5\%$$

What does statistical theory say?

The answers to these exercises are on p. A84.

5. THE GALLUP POLL

The Gallup Poll does not use a simple random sample (section 4 of chapter 19). As a result, they do not estimate their standard errors using the method of this chapter. However, it is interesting to compare their samples to simple random samples of the same size. For instance, in 1952 they predicted a 51% vote for Eisenhower, based on a sample of 5,385 people. With a simple random sample,

$$\text{SE for number} = \sqrt{5{,}385} \times \sqrt{0.51 \times 0.49} \approx 37$$

$$\text{SE for percent} = \frac{37}{5{,}385} \times 100\% \approx 0.7 \text{ of } 1\%.$$

In fact, Eisenhower got 54.9% of the vote in that election. The Gallup Poll estimate was off by 3.9 percentage points. This is nearly 6 times the SE for a simple random sample. Table 1 shows the comparison for every presidential election from 1952 to 2004.

Table 1. Comparing the Gallup Poll with a simple random sample. The errors of prediction are on the whole quite a bit bigger than those to be expected from a simple random sample of the same size.

Year	*Sample size*	*SE for simple random sample*	*Actual error*
1952	5,385	0.7 of 1%	3.9%
1956	8,144	0.5 of 1%	2.1%
1960	8,015	0.6 of 1%	1.3%
1964	6,625	0.6 of 1%	2.9%
1968	4,414	0.7 of 1%	0.4 of 1%
1972	3,689	0.8 of 1%	1.8%
1976	3,439	0.9 of 1%	2.0%
1980	3,500	0.8 of 1%	3.5%
1984	3,456	0.8 of 1%	0.5 of 1%
1988	4,089	0.8 of 1%	2.9%
1992	2,019	1.1%	6.1%
1996	2,895	0.9%	2.8%
2000	3,571	0.8 of 1%	0.2%
2004	2,014	1.1%	1.6%

Source: See table 4 in chapter 19.

In 11 elections out of 14, the error was considerably larger than the SE for a simple random sample. One reason is that predictions are based only on part of the sample, namely, those people judged likely to vote (section 6 of chapter 19).

This eliminates about half the sample. Table 2 compares the errors made by the Gallup Poll with SEs computed for simple random samples whose size equals the number of likely voters. The simple random sample formula is still not doing a good job at predicting the size of the errors.

Why not? Well, the Gallup Poll is not drawing tickets at random from a box—although the telephone samples used from 1992 onwards come closer to simple random sampling than designs used before that (pp. 340–341, 346). Three other issues should be mentioned: (i) the process used to screen out the non-voters may break down at times; (ii) some voters may still not have decided how to vote when they are interviewed; (iii) voters may change their minds between the last pre-election poll and election day, especially in close contests. In a volatile, three-way contest like the 1992 election, such problems take their toll (p. 346).

Table 2. The accuracy of the Gallup Poll compared to that of a simple random sample whose size equals the number of likely voters in the Gallup Poll sample.

Year	*Number of likely voters*	*SE for simple random sample*	*Actual error*
1952	3,350	0.9 of 1%	3.9%
1956	4,950	0.7 of 1%	2.1%
1960	5,100	0.7 of 1%	1.3%
1964	4,100	0.8 of 1%	2.9%
1968	2,700	1.0%	0.4 of 1%
1972	2,100	1.1%	1.8%
1976	2,000	1.1%	2.0%
1980	2,000	1.1%	3.5%
1984	2,000	1.1%	0.5 of 1%
1988	2,600	1.0%	2.9%
1992	1,600	1.2%	6.1%
1996	1,100	1.5%	2.8%
2000	2,400	1.0%	0.2%
2004	1,600	1.2%	1.6%

Note: The number of likely voters is rounded.
Source: The Gallup Poll (American Institute of Public Opinion).

Exercise Set E

1. A Gallup Poll pre-election survey based on a sample of 1,000 people estimates a 65% vote for the Democratic candidate in a certain election. True or false, and explain: the likely size of the chance error in this estimate can be figured as follows—

$$\sqrt{1{,}000} \times \sqrt{0.65 \times 0.35} \approx 15, \qquad \frac{15}{1{,}000} \times 100\% = 1.5\%$$

2. One thousand tickets are drawn at random without replacement from a large box, and 651 of the draws show a 1. The fraction of 1's in the box is estimated as 65%. True or false, and explain: the likely size of the chance error in this estimate can be figured as follows—

$$\sqrt{1{,}000} \times \sqrt{0.65 \times 0.35} \approx 15, \qquad \frac{15}{1{,}000} \times 100\% = 1.5\%$$

3. The following article appeared on the *New York Times* Op Ed page of August 27, 1988, headlined MAYBE BUSH HAS ALREADY WON.

   The presidential campaign, only now formally set to begin, is in fact virtually finished. Despite the Niagara of news stories about how the candidates are touting their running mates, haggling over debates and sniping at each other, the die is just about cast.

   A significant indicator is the Gallup Poll, which this week shows Vice President Bush ahead of Gov. Michael S. Dukakis by 4 percentage points. In the half century since George Gallup began his electoral opinion surveys in Presidential years, his "trial heats" in the last week or so of September have foretold with notable accuracy the outcome on election day.

   The late James A. Farley, the Democrats' peerless tactician of 50 years ago, always argued that voters made up their minds by Labor Day. . . . It is now established, moreover, that when traditional nonvoters—the object of get-out-the-vote efforts—are persuaded to vote, they too cast their ballots in the same proportion as the rest of the electorate . . . . Significant changes in the percentages from September to November are due only to altered voter enthusiasm . . . .

   (a) How does the article explain differences in voter opinion between September and November?
   (b) What else could explain a difference between Gallup Poll results in late September and election results in early November?
   (c) A difference of several percentage points between Gallup Poll results in late September and election results in early November is: very unlikely, unlikely but possible, quite possible. Choose one option, and explain.

The answers to these exercises are on p. A84.

6. REVIEW EXERCISES

Review exercises may cover material from previous chapters.

1. A survey organization draws a simple random sample of 1,000 registered voters in a certain town. In the sample, 32% approve of the Mayor. The organization estimates that 32% of all 50,000 registered voters in the town approve of the Mayor. How to figure the SE? The organization realizes that the number in the sample who approve _____ 1,000 draws _____ box _____. Fill in each blank (33 words or less). Then work out the SE.

2. The Residential Energy Consumption Survey found in 2001 that 47% of American households had internet access.[10] A market survey organization repeated this study in a certain town with 25,000 households, using a simple random sample of 500 households: 239 of the sample households had internet access.

   (a) The percentage of households in the town with internet access is estimated as _____; this estimate is likely to be off by _____ or so.
   (b) If possible, find a 95%-confidence interval for the percentage of all 25,000 households with internet access. If this is not possible, explain why not.

3. Of the 500 sample households in the previous exercise, 7 had three or more large-screen TVs.
   (a) The percentage of households in the town with three or more large-screen TVs is estimated as _____; this estimate is likely to be off by _____ or so.
   (b) If possible, find a 95%-confidence interval for the percentage of all 25,000 households with three or more large-screen TVs. If this is not possible, explain why not.

4. (This continues exercise 3.) Among the sample households, 121 had no car, 172 had one car, and 207 had two or more cars. Estimate the percentage of households in the town with one or more cars; attach a standard error to the estimate. If this is not possible, explain why not.

5. The National Assessment of Educational Progress administers standardized achievement tests to nationwide samples of 17-year-olds in school. One year, the tests covered history and literature. You may assume that a simple random sample of size 6,000 was taken. Only 36.1% of the students in the sample knew that Chaucer wrote *The Canterbury Tales*, but 95.2% knew that Edison invented the light bulb.[11]
   (a) If possible, find a 95%-confidence interval for the percentage of all 17-year-olds in school who knew that Chaucer wrote *The Canterbury Tales*. If this is not possible, why not?
   (b) If possible, find a 95%-confidence interval for the percentage of all 17-year-olds in school who knew that Edison invented the light bulb. If this is not possible, why not?

6. True or false: with a well-designed sample survey, the sample percentage is very likely to equal the population percentage. Explain.

7. (Hypothetical.) One year, there were 252 trading days on the New York Stock Exchange, and IBM common stock went up on 131 of them: $131/252 \approx 52\%$. A statistician attaches a standard error to this percentage as follows:

$$\text{SE for number} = \sqrt{252} \times \sqrt{0.52 \times 0.48} \approx 8$$

$$\text{SE for percent} = \frac{8}{252} \times 100\% \approx 3\%$$

   Is this the right SE? Answer yes or no, and explain.

8. A simple random sample of 3,500 people age 18 or over is taken in a large town to estimate the percentage of people (age 18 and over in that town) who read newspapers. It turns out that 2,487 people in the sample are newspaper readers.[12] The population percentage is estimated as

$$\frac{2{,}487}{3{,}500} \times 100\% \approx 71\%$$

   The standard error is estimated as 0.8 of 1%, because

$$\sqrt{3{,}500} \times \sqrt{0.71 \times 0.29} \approx 27, \qquad \frac{27}{3{,}500} \times 100\% \approx 0.8 \text{ of } 1\%$$

(a) Is 0.8 of 1% the right SE? Answer yes or no, and explain.
(b) 71% ± 1.6% is a _____ for the _____. Fill in the blanks and explain.

9. (Hypothetical.) A bank wants to estimate the amount of change people carry. They take a simple random sample of 100 people, and find that on the average, people in the sample carry 73¢ in change. They figure the standard error is 4¢, because

$$\sqrt{100} \times \sqrt{0.73 \times 0.27} \approx 4, \quad 4/100 = .04$$

Are they right? Answer yes or no, and explain.

10. In Keno, there are 80 balls numbered from 1 to 80, and 20 are drawn at random. If you play a double-number, you win if both numbers are chosen. This bet pays 11 to 1, and you have very close to a 6% chance of winning.[13] If you play 100 times and stake $1 on a double-number each time, your net gain will be around _____, give or take _____ or so.

11. One hundred draws will be made at random without replacement from a large box of numbered tickets. There are two options:

(i) To win $1 if the sum of the draws is bigger than 710.
(ii) To win $1 if the average of the draws is bigger than 7.1.

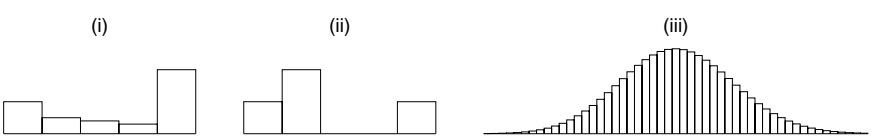
Which is better? Or are they the same? Explain.

12. A monthly opinion survey is based on a sample of 1,500 persons, "scientifically chosen as a representative cross section of the American public." The press release warns that the estimates are subject to chance error, but guarantees that they are "reliable to within two percentage points." The word "reliable" is ambiguous. According to statistical theory, the guarantee should be interpreted as follows:

(i) In virtually all these surveys, the estimates will be within two percentage points of the parameters.
(ii) In most such surveys, the estimates will be within two percentage points of the parameters, but in some definite percentage of the time larger errors are expected.

Explain.

13. One hundred draws are made at random with replacement from the box $\boxed{1\ 2\ 2\ 5}$. One of the graphs below is a histogram for the numbers drawn. Another is the probability histogram for the sum. And the third is irrelevant. Which is which? Why?



14. A coin is tossed 1,000 times.

(a) Suppose it lands heads 529 times. Find the expected value for the number of heads, the chance error, and the standard error.

(b) Suppose it lands heads 484 times. Find the expected value for the number of heads, the chance error, and the standard error.
(c) Suppose it lands heads 514 times. Find the expected value for the number of heads, the chance error, and the standard error.

15. A survey organization takes a simple random sample of 1,500 persons from the residents of a large city. Among these sample persons, 1,035 were renters.

(a) The expected value for the percentage of sample persons who rent is _____ 69%.
(b) The SE for the percentage of sample persons who rent is _____ 1.2%.

Fill in the blanks, and explain. Options:

(i) exactly equal to (ii) estimated from the data as

7. SUMMARY

1. With a simple random sample, the sample percentage is used to estimate the population percentage.

2. The sample percentage will be off the population percentage, due to chance error. The SE for the sample percentage tells you the likely size of the amount off.

3. When sampling from a 0–1 box whose composition is unknown, the SD of the box can be estimated by substituting the fractions of 0's and 1's in the sample for the unknown fractions in the box. This *bootstrap estimate* is good when the sample is large.

4. A *confidence interval* for the population percentage is obtained by going the right number of SEs either way from the sample percentage. The confidence level is read off the normal curve. This method should only be used with large samples.

5. In the frequency theory of probability, parameters are not subject to chance variation. That is why confidence statements are made instead of probability statements.

6. The formulas for simple random samples may not apply to other kinds of samples. If the sample was not chosen by a probability method, watch out: SEs computed from the formulas may not mean very much.